

Problem

Determine the inverse Fourier transforms of the following:

(a) $F(\omega) = \frac{e^{-j2\omega}}{1 + j\omega}$

(b) $H(\omega) = \frac{1}{(j\omega + 4)^2}$

(c) $G(\omega) = 2u(\omega + 1) - 2u(\omega - 1)$

Step-by-step solution

Step 1 of 3

(A) $f(t) = e^{-(t-2)}u(t)$

Step 2 of 3

(B) $h(t) = t e^{-at}u(t)$

Step 3 of 3

(C) if $x(t) = u(t+1) - u(t-1) \rightarrow X(\omega) = \frac{2 \sin \omega}{\omega}$

$G(\omega) = 2u(\omega+1) - 2u(\omega-1) \rightarrow g(t) = \frac{2 \sin t}{\pi t}$